

Variety of life

LIFE ON EARTH IS ORGANIZED INTO RELATED GROUPS.

Scientists have attempted to make sense of Earth's biodiversity—its enormous variety of life—by classifying living things into different groups, according to how they look and how they are related.

Three domains of life

Biologists estimate that there are about eight million species of living things on Earth today. The field of biology that organizes all these species into an understandable system is called taxonomy. Taxonomy arranges organisms in a hierarchy of groups. The largest groups are called domains. Most biologists divide life into three domains: Bacteria, Eukaryota, and the Archaea.

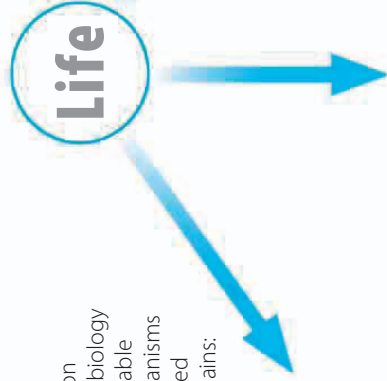


Bacteria

These simple-celled organisms live in all parts of Earth, from deep inside rocks to the guts of most eukaryotes. A few bacteria infect eukaryotes, causing diseases.

Eukaryota

This domain includes plants, animals, fungi, and some single-celled organisms. The Eukaryota is the only domain to contain multicellular organisms, where body cells work together to do different jobs.



The word “dolphin” means “womb fish”—early biologists thought dolphins were related to fish, and not land mammals.

Archaea come in many different shapes, from strands like these, to cubes, and even spherical varieties



Archaea

These are the oldest living things on Earth. They evolved more than 3.8 billion years ago out of the extreme conditions on Earth back then, and can still be found today in conditions too harsh for other life.

Classification

Taxonomists group organisms according to how they are related to each other. Group members have all evolved from a common ancestor at some point in the past. The further you go down the groups, the closer the similarities are between species.

▷ KINGDOM

Eukaryota is the largest domain and it is the only one that is subdivided into kingdoms.

Animal kingdom

Every animal belongs to this group. They all have multicellular bodies, must feed on other organisms to survive, and are usually able to respond rapidly to threats and problems.

Fungi kingdom

Until the middle of the 20th century, these organisms were considered a branch of the plant kingdom. Fungi are molds and mushrooms that live in damp habitats, and grow on their food, which they break down and absorb outside themselves.

Protist kingdom

The protists are a diverse group of eukaryotes that do not develop into specialized multicellular bodies. Instead they survive as single, solitary cells. However, a few species develop into clusters or colonies of individual cells.

Plant kingdom

Plants are multicellular organisms that make their own food by photosynthesis. Most plants are terrestrial or live in freshwater, and live in one place during their lifetime, although they can move in response to their environment.

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Fish, amphibians, and reptiles **58–59 >**

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